

Code-Layout, Readability and Reusability

Date	4 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Standard 4-space indentation used.
2	Proper File Structure	Files and folders are logically organized	Yes	Separated by app, model, and static folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive names like hdi_score, gni_input.
4	Function / Method Names	Functions are descriptively named	Yes	Functions like calculate_hdi() used.
5	Code Comments	Inline and block comments explain logic	Yes	Documented all core ML logic.
6	Modular Design	Code is split into reusable functions/modules	Yes	Model logic separated from API routes.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Cleaned up all temporary testing code.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Added try-except blocks for API inputs.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Script	Python (Pandas)	Used in Model Training & Testing	High

2	HDI Prediction Model	Scikit-learn (Pickle)	Used in API Prediction Route	High
3	UI Components (CSS)	Bootstrap / CSS	Used across all HTML pages	High
4	Utility Helper Functions	Python	Used in Data validation	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clean and logical folder architecture.
Readability	5	Well-commented code with clear naming.
Reusability	4	Modular code allows for easy extensions.
Documentation / Comments	5	All functions are well-documented.
Overall Score	4.45/5	Excellent maintainable codebase.